

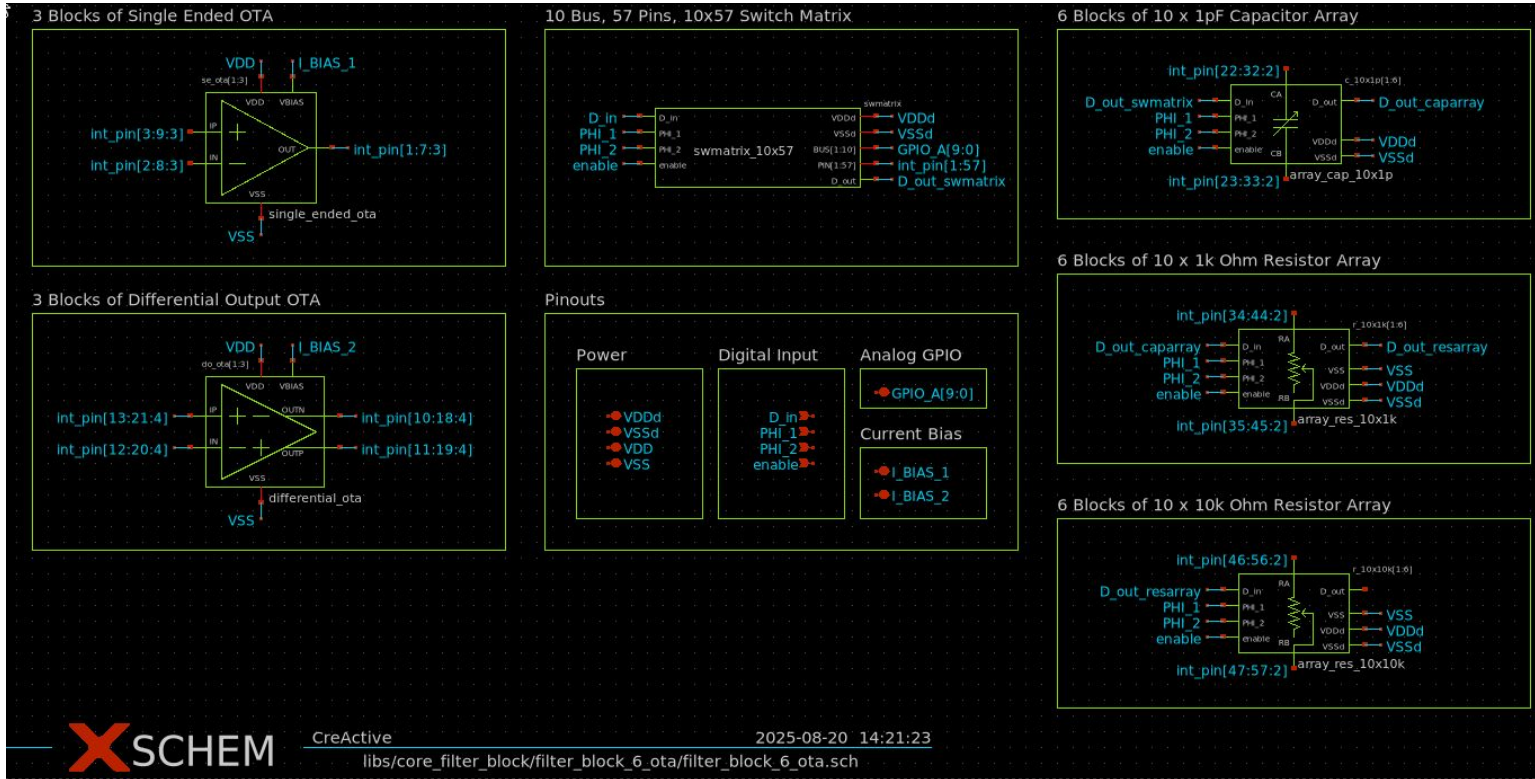
M3: CreActive

Active filter building blocks

Schematic & Simulation Review

Refer to [this link](#) for more updated result

Top Level Schematic

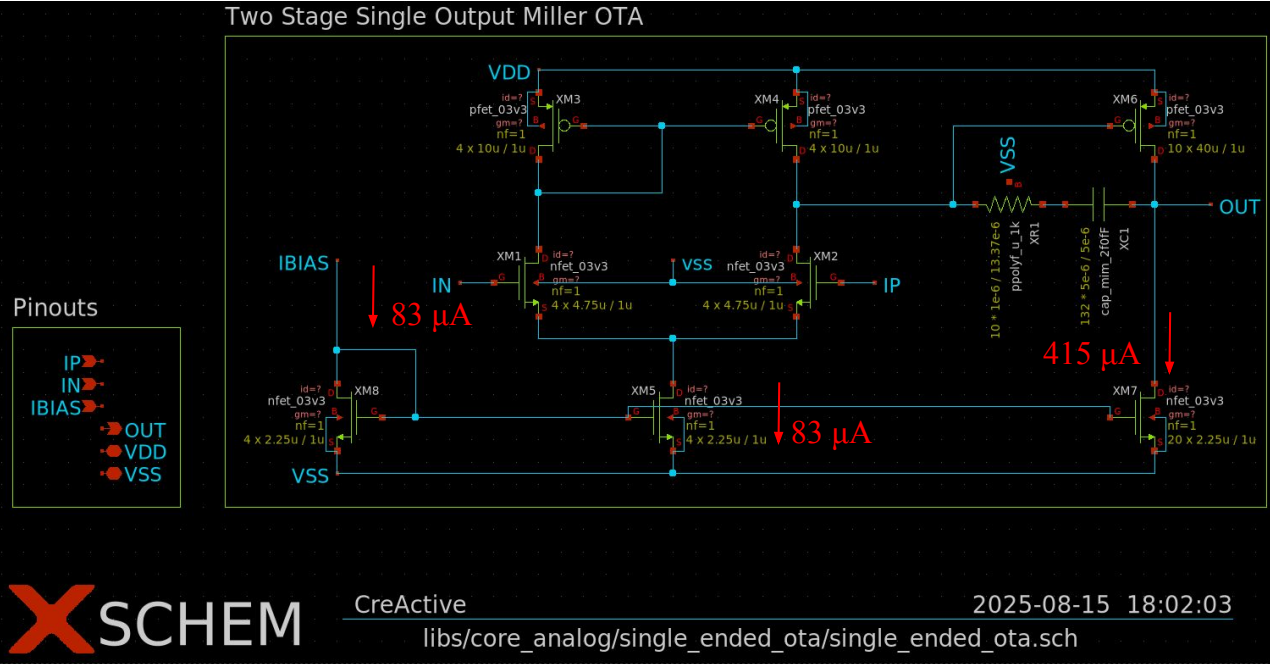


- All of the building blocks (single output OTA, differential output OTA, Capacitor Array, and Resistor array) have been completed

Top Level Simulation

- The simulation was done with the switch matrix connected to op amp terminals (adding more parasitic resistance and capacitance)
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Two Stage Single Output Miller OTA: (Schematic)



Parameter	Value
C load	30 pF
C miller	6.6 pF
Rz (Nulling Resistor)	1337 Ω

Sizing script (using gm/id): [ota-sizing](#)

Sizing reference:

- CMOS Analog Circuit Design, Philip E. Allen
- Analog Circuit Design, Harald Pertl
- Systematic Circuit Design

Two Stage Single Output Miller OTA: (Testbench)

MODELS

```
.include $::180MCU_MODELS/design.ngspice
lib $::180MCU_MODELS/sml41064.ngspice typical
lib $::180MCU_MODELS/sml41064.ngspice cap_mim
lib $::180MCU_MODELS/sml41064.ngspice res_typical
lib $::180MCU_MODELS/sml41064.ngspice moscap_typical
lib $::180MCU_MODELS/sml41064.ngspice mimcap_typical
* .lib $::180MCU_MODELS/sml41064.ngspice res_statistical
```

s1

```
.option wnflag=1
.option safecurrents
.option solver=klu
```

```
vin1 in1 gnd pulse(0 3.3 0 10n 10n 5u 10u)
.control
reset
save all
set num_threads=8
op
show
write tb_single_ended_ota.raw
*dc vin1 -1 4 0.1
*tran 10n 30u
ac dec 100 1 100e9
```

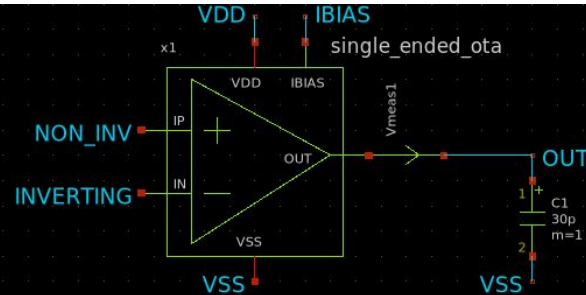
```

let vout_mag =abs(v(out))
let vout_phase_margin = phase(v(out))*180/pi + 180
meas ac Aol find vout_mag at = 10
meas ac UGF when vout_mag=1 fall=1
meas ac PM find vout_phase_margin when vout_mag=1

write tb_single_ended_ota.raw

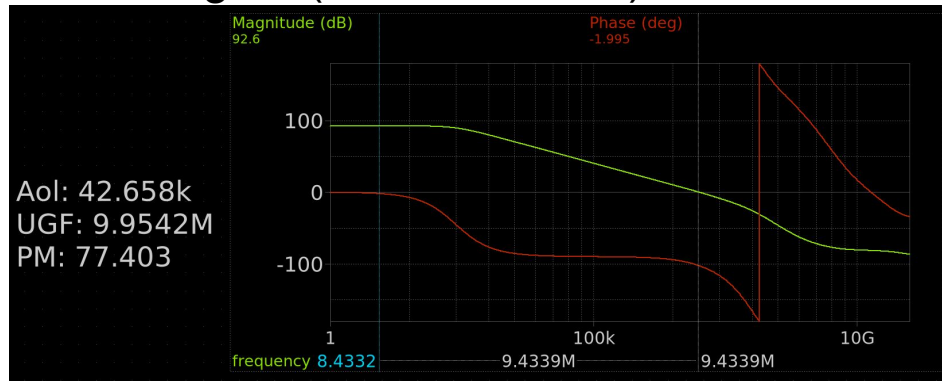
.endc

```



Two Stage Single Output Miller OTA: (Simulation)

Small Signal (AC Simulation)

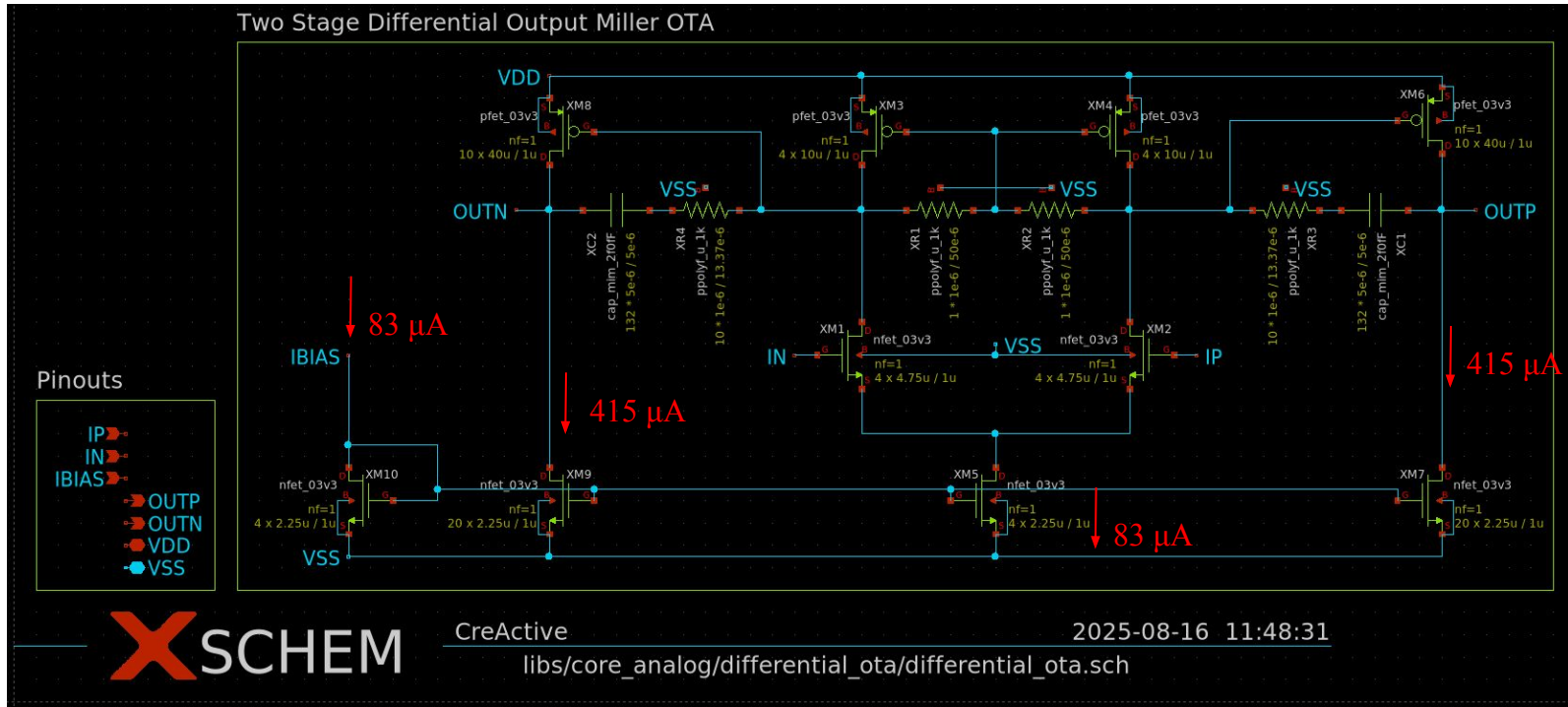


Parameter	Value
DC gain	93.94 dB
Unity Gain Frequency	8.53 MHz
Phase Margin	75.55 deg



Parameter	Value
ICMR	0.4847 V - 3.221 V
Slew Rate	18.12 V/us
Settling time	0.26 us

Two Stage Differential Output Miller OTA: (Schematic)

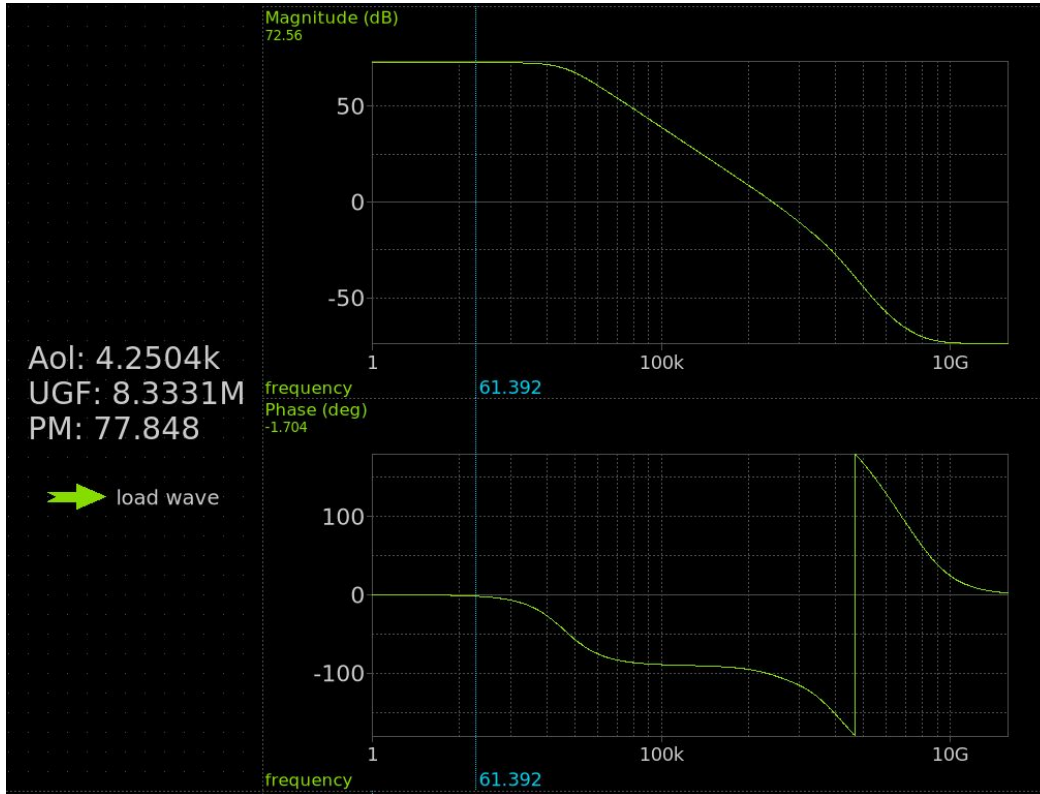


- The sizing of this differential output OTA used the same method as the single output OTA, but there is a slight difference in the topology. There are two resistor (R1 & R2) in the input stage to provide bias voltage to the active load (M3 & M4). The common source output stage from the non-inverting side is mirrored to the inverting side to provide a differential output.

Two Stage Differential Output Miller OTA: (Testbench)

Two Stage Differential Output Miller OTA: (Simulation)

AC Simulation



Parameter	Value
DC gain	72.56 dB
Unity Gain Frequency	8.33 MHz
Phase Margin	77.848 deg

Two Stage Differential Output Miller OTA: (Simulation)

Tran Simulation

Parameter	Value
DC gain	65.186 dB
Unity Gain Frequency	3.42 MHz
Phase Margin	88.424 deg

Sallen Key 2nd Order Active Filter Testing